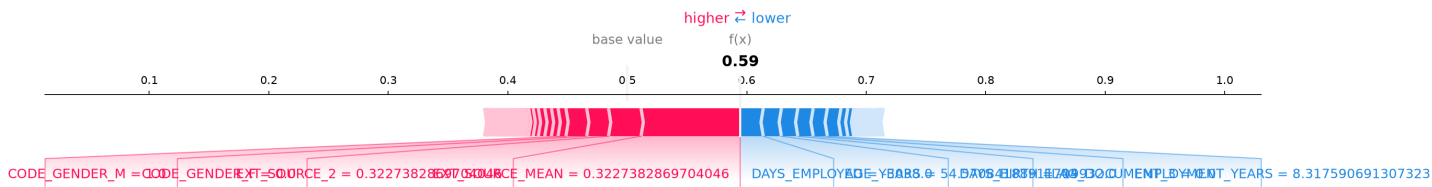


Local Interpretability Report

Customer ID: 100007

Prediction Score: 1.0000

SHAP Force Plot



Client Data vs. Population Average

| Feature                    | Client Value | Dataset Average |
|----------------------------|--------------|-----------------|
| CNT_CHILDREN               | 0.00         | 0.42            |
| AMT_INCOME_TOTAL           | 121500.00    | 168797.92       |
| AMT_CREDIT                 | 513000.00    | 599026.00       |
| AMT_ANNUITY                | 21865.50     | 27108.49        |
| AMT_GOODS_PRICE            | 513000.00    | 538316.29       |
| REGION_POPULATION_RELATIVE | 0.03         | 0.02            |
| DAYS_BIRTH                 | -19932.00    | -16037.00       |
| DAYS_EMPLOYED              | -3038.00     | -2251.61        |
| DAYS_REGISTRATION          | -4311.00     | -4986.12        |
| DAYS_ID_PUBLISH            | -3458.00     | -2994.20        |
| FLAG_MOBIL                 | 1.00         | 1.00            |
| FLAG_EMP_PHONE             | 1.00         | 0.82            |

|                             |       |       |
|-----------------------------|-------|-------|
| FLAG_WORK_PHONE             | 0.00  | 0.20  |
| FLAG_CONT_MOBILE            | 1.00  | 1.00  |
| FLAG_PHONE                  | 0.00  | 0.28  |
| FLAG_EMAIL                  | 0.00  | 0.06  |
| CNT_FAM_MEMBERS             | 1.00  | 2.15  |
| REGION_RATING_CLIENT        | 2.00  | 2.05  |
| REGION_RATING_CLIENT_W_CITY | 2.00  | 2.03  |
| HOUR_APPR_PROCESS_START     | 11.00 | 12.06 |
| REG_REGION_NOT_LIVE_REGION  | 0.00  | 0.02  |
| REG_REGION_NOT_WORK_REGION  | 0.00  | 0.05  |
| LIVE_REGION_NOT_WORK_REGION | 0.00  | 0.04  |
| REG_CITY_NOT_LIVE_CITY      | 0.00  | 0.08  |
| REG_CITY_NOT_WORK_CITY      | 1.00  | 0.23  |
| LIVE_CITY_NOT_WORK_CITY     | 1.00  | 0.18  |
| EXT_SOURCE_1                | 0.51  | 0.50  |
| EXT_SOURCE_2                | 0.32  | 0.51  |
| EXT_SOURCE_3                | 0.54  | 0.52  |
| APARTMENTS_AVG              | 0.09  | 0.10  |
| BASEMENTAREA_AVG            | 0.08  | 0.08  |
| YEARS_BEGINEXPLUATATION_AVG | 0.98  | 0.98  |
| ELEVATORS_AVG               | 0.00  | 0.04  |
| ENTRANCES_AVG               | 0.14  | 0.14  |

|                              |      |      |
|------------------------------|------|------|
| FLOORSMAX_AVG                | 0.17 | 0.20 |
| LANDAREA_AVG                 | 0.05 | 0.06 |
| LIVINGAREA_AVG               | 0.07 | 0.09 |
| NONLIVINGAREA_AVG            | 0.00 | 0.01 |
| APARTMENTS_MODE              | 0.08 | 0.10 |
| BASEMENTAREA_MODE            | 0.07 | 0.08 |
| YEARS_BEGINEXPLUATATION_MODE | 0.98 | 0.98 |
| ELEVATORS_MODE               | 0.00 | 0.03 |
| ENTRANCES_MODE               | 0.14 | 0.14 |
| FLOORSMAX_MODE               | 0.17 | 0.19 |
| LANDAREA_MODE                | 0.05 | 0.05 |
| LIVINGAREA_MODE              | 0.07 | 0.09 |
| NONLIVINGAREA_MODE           | 0.00 | 0.01 |
| APARTMENTS_MEDI              | 0.09 | 0.10 |
| BASEMENTAREA_MEDI            | 0.08 | 0.08 |
| YEARS_BEGINEXPLUATATION_MEDI | 0.98 | 0.98 |
| ELEVATORS_MEDI               | 0.00 | 0.04 |
| ENTRANCES_MEDI               | 0.14 | 0.14 |
| FLOORSMAX_MEDI               | 0.17 | 0.20 |
| LANDAREA_MEDI                | 0.05 | 0.06 |
| LIVINGAREA_MEDI              | 0.07 | 0.09 |
| NONLIVINGAREA_MEDI           | 0.00 | 0.01 |

|                          |          |         |
|--------------------------|----------|---------|
| TOTALAREA_MODE           | 0.07     | 0.09    |
| OBS_30_CNT_SOCIAL_CIRCLE | 0.00     | 1.42    |
| DEF_30_CNT_SOCIAL_CIRCLE | 0.00     | 0.14    |
| OBS_60_CNT_SOCIAL_CIRCLE | 0.00     | 1.40    |
| DEF_60_CNT_SOCIAL_CIRCLE | 0.00     | 0.10    |
| DAYS_LAST_PHONE_CHANGE   | -1106.00 | -962.86 |
| FLAG_DOCUMENT_2          | 0.00     | 0.00    |
| FLAG_DOCUMENT_3          | 0.00     | 0.71    |
| FLAG_DOCUMENT_4          | 0.00     | 0.00    |
| FLAG_DOCUMENT_5          | 0.00     | 0.02    |
| FLAG_DOCUMENT_6          | 0.00     | 0.09    |
| FLAG_DOCUMENT_7          | 0.00     | 0.00    |
| FLAG_DOCUMENT_8          | 1.00     | 0.08    |
| FLAG_DOCUMENT_9          | 0.00     | 0.00    |
| FLAG_DOCUMENT_10         | 0.00     | 0.00    |
| FLAG_DOCUMENT_11         | 0.00     | 0.00    |
| FLAG_DOCUMENT_12         | 0.00     | 0.00    |
| FLAG_DOCUMENT_13         | 0.00     | 0.00    |
| FLAG_DOCUMENT_14         | 0.00     | 0.00    |
| FLAG_DOCUMENT_15         | 0.00     | 0.00    |
| FLAG_DOCUMENT_16         | 0.00     | 0.01    |
| FLAG_DOCUMENT_17         | 0.00     | 0.00    |

|                                       |           |          |
|---------------------------------------|-----------|----------|
| FLAG_DOCUMENT_18                      | 0.00      | 0.01     |
| FLAG_DOCUMENT_19                      | 0.00      | 0.00     |
| FLAG_DOCUMENT_20                      | 0.00      | 0.00     |
| FLAG_DOCUMENT_21                      | 0.00      | 0.00     |
| AMT_REQ_CREDIT_BUREAU_HOUR            | 0.00      | 0.01     |
| AMT_REQ_CREDIT_BUREAU_DAY             | 0.00      | 0.01     |
| AMT_REQ_CREDIT_BUREAU_WEEK            | 0.00      | 0.03     |
| AMT_REQ_CREDIT_BUREAU_MON             | 0.00      | 0.23     |
| AMT_REQ_CREDIT_BUREAU_QRT             | 0.00      | 0.23     |
| AMT_REQ_CREDIT_BUREAU_YEAR            | 0.00      | 1.78     |
| DAYS_EMPLOYED_ANOMALOUS               | 0.00      | 0.18     |
| AGE_YEARS                             | 54.57     | 43.91    |
| EMPLOYMENT_YEARS                      | 8.32      | 6.16     |
| CREDIT_INCOME_RATIO                   | 4.22      | 3.96     |
| ANNUITY_INCOME_RATIO                  | 0.18      | 0.18     |
| CREDIT_ANNUITY_RATIO                  | 23.46     | 21.61    |
| INCOME_PER_PERSON                     | 121500.00 | 93105.76 |
| EXT_SOURCE_MEAN                       | 0.32      | 0.51     |
| NAME_CONTRACT_TYPE_Cash<br>loans      | 1.00      | 0.90     |
| NAME_CONTRACT_TYPE_Revolving<br>loans | 0.00      | 0.10     |
| CODE_GENDER_F                         | 0.00      | 0.66     |

|                                          |      |      |
|------------------------------------------|------|------|
| CODE_GENDER_M                            | 1.00 | 0.34 |
| CODE_GENDER_XNA                          | 0.00 | 0.00 |
| FLAG_OWN_CAR_N                           | 1.00 | 0.66 |
| FLAG_OWN_CAR_Y                           | 0.00 | 0.34 |
| FLAG_OWN_REALTY_N                        | 0.00 | 0.31 |
| FLAG_OWN_REALTY_Y                        | 1.00 | 0.69 |
| NAME_TYPE_SUITE_Children                 | 0.00 | 0.01 |
| NAME_TYPE_SUITE_Family                   | 0.00 | 0.13 |
| NAME_TYPE_SUITE_Group of<br>people       | 0.00 | 0.00 |
| NAME_TYPE_SUITE_Missing                  | 0.00 | 0.00 |
| NAME_TYPE_SUITE_Other_A                  | 0.00 | 0.00 |
| NAME_TYPE_SUITE_Other_B                  | 0.00 | 0.01 |
| NAME_TYPE_SUITE_Spouse,<br>partner       | 0.00 | 0.04 |
| NAME_TYPE_SUITE_Unaccompanied            | 1.00 | 0.81 |
| NAME_INCOME_TYPE_Businessman             | 0.00 | 0.00 |
| NAME_INCOME_TYPE_Commercial<br>associate | 0.00 | 0.23 |
| NAME_INCOME_TYPE_Maternity<br>leave      | 0.00 | 0.00 |
| NAME_INCOME_TYPE_Pensioner               | 0.00 | 0.18 |
| NAME_INCOME_TYPE_State<br>servant        | 0.00 | 0.07 |
| NAME_INCOME_TYPE_Student                 | 0.00 | 0.00 |

|                                                   |      |      |
|---------------------------------------------------|------|------|
| NAME_INCOME_TYPE_Unemployed                       | 0.00 | 0.00 |
| NAME_INCOME_TYPE_Working                          | 1.00 | 0.52 |
| NAME_EDUCATION_TYPE_Academic degree               | 0.00 | 0.00 |
| NAME_EDUCATION_TYPE_Higher education              | 0.00 | 0.24 |
| NAME_EDUCATION_TYPE_Incomplete higher             | 0.00 | 0.03 |
| NAME_EDUCATION_TYPE_Lower secondary               | 0.00 | 0.01 |
| NAME_EDUCATION_TYPE_Secondary / secondary special | 1.00 | 0.71 |
| NAME_FAMILY_STATUS_Civil marriage                 | 0.00 | 0.10 |
| NAME_FAMILY_STATUS_Married                        | 0.00 | 0.64 |
| NAME_FAMILY_STATUS_Separated                      | 0.00 | 0.06 |
| NAME_FAMILY_STATUS_Single / not married           | 1.00 | 0.15 |
| NAME_FAMILY_STATUS_Unknown                        | 0.00 | 0.00 |
| NAME_FAMILY_STATUS_Widow                          | 0.00 | 0.05 |
| NAME_HOUSING_TYPE_Co-op apartment                 | 0.00 | 0.00 |
| NAME_HOUSING_TYPE_House / apartment               | 1.00 | 0.89 |
| NAME_HOUSING_TYPE_Municipal apartment             | 0.00 | 0.04 |
| NAME_HOUSING_TYPE_Office apartment                | 0.00 | 0.01 |
| NAME_HOUSING_TYPE_Rented apartment                | 0.00 | 0.02 |

|                                       |      |      |
|---------------------------------------|------|------|
| NAME_HOUSING_TYPE_With parents        | 0.00 | 0.05 |
| OCCUPATION_TYPE_Accountants           | 0.00 | 0.03 |
| OCCUPATION_TYPE_Cleaning staff        | 0.00 | 0.02 |
| OCCUPATION_TYPE_Cooking staff         | 0.00 | 0.02 |
| OCCUPATION_TYPE_Core staff            | 1.00 | 0.09 |
| OCCUPATION_TYPE_Drivers               | 0.00 | 0.06 |
| OCCUPATION_TYPE_HR staff              | 0.00 | 0.00 |
| OCCUPATION_TYPE_High skill tech staff | 0.00 | 0.04 |
| OCCUPATION_TYPE_IT staff              | 0.00 | 0.00 |
| OCCUPATION_TYPE_Laborers              | 0.00 | 0.18 |
| OCCUPATION_TYPE_Low-skill Laborers    | 0.00 | 0.01 |
| OCCUPATION_TYPE_Managers              | 0.00 | 0.07 |
| OCCUPATION_TYPE_Medicine staff        | 0.00 | 0.03 |
| OCCUPATION_TYPE_Missing               | 0.00 | 0.31 |
| OCCUPATION_TYPE_Private service staff | 0.00 | 0.01 |
| OCCUPATION_TYPE_Realty agents         | 0.00 | 0.00 |
| OCCUPATION_TYPE_Sales staff           | 0.00 | 0.10 |
| OCCUPATION_TYPE_Secretaries           | 0.00 | 0.00 |
| OCCUPATION_TYPE_Security staff        | 0.00 | 0.02 |
| OCCUPATION_TYPE_Waiters/barmen staff  | 0.00 | 0.00 |

|                                             |      |      |
|---------------------------------------------|------|------|
| WEEKDAY_APPR_PROCESS_START_FRIDAY           | 0.00 | 0.16 |
| WEEKDAY_APPR_PROCESS_START_MONDAY           | 0.00 | 0.16 |
| WEEKDAY_APPR_PROCESS_START_SATURDAY         | 0.00 | 0.11 |
| WEEKDAY_APPR_PROCESS_START_SUNDAY           | 0.00 | 0.05 |
| WEEKDAY_APPR_PROCESS_START_THURSDAY         | 1.00 | 0.16 |
| WEEKDAY_APPR_PROCESS_START_TUESDAY          | 0.00 | 0.18 |
| WEEKDAY_APPR_PROCESS_START_WEDNESDAY        | 0.00 | 0.17 |
| ORGANIZATION_TYPE_Advertising               | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Agriculture               | 0.00 | 0.01 |
| ORGANIZATION_TYPE_Bank                      | 0.00 | 0.01 |
| ORGANIZATION_TYPE_Business<br>Entity Type 1 | 0.00 | 0.02 |
| ORGANIZATION_TYPE_Business<br>Entity Type 2 | 0.00 | 0.03 |
| ORGANIZATION_TYPE_Business<br>Entity Type 3 | 0.00 | 0.22 |
| ORGANIZATION_TYPE_Cleaning                  | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Construction              | 0.00 | 0.02 |
| ORGANIZATION_TYPE_Culture                   | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Electricity               | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Emergency                 | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Government                | 0.00 | 0.03 |
| ORGANIZATION_TYPE_Hotel                     | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Housing                   | 0.00 | 0.01 |

|                                        |      |      |
|----------------------------------------|------|------|
| ORGANIZATION_TYPE_Industry:<br>type 1  | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Industry:<br>type 10 | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Industry:<br>type 11 | 0.00 | 0.01 |
| ORGANIZATION_TYPE_Industry:<br>type 12 | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Industry:<br>type 13 | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Industry:<br>type 2  | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Industry:<br>type 3  | 0.00 | 0.01 |
| ORGANIZATION_TYPE_Industry:<br>type 4  | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Industry:<br>type 5  | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Industry:<br>type 6  | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Industry:<br>type 7  | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Industry:<br>type 8  | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Industry:<br>type 9  | 0.00 | 0.01 |
| ORGANIZATION_TYPE_Insurance            | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Kindergarten         | 0.00 | 0.02 |
| ORGANIZATION_TYPE_Legal<br>Services    | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Medicine             | 0.00 | 0.04 |

|                                          |      |      |
|------------------------------------------|------|------|
| ORGANIZATION_TYPE_Military               | 0.00 | 0.01 |
| ORGANIZATION_TYPE_Mobile                 | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Other                  | 0.00 | 0.05 |
| ORGANIZATION_TYPE_Police                 | 0.00 | 0.01 |
| ORGANIZATION_TYPE_Postal                 | 0.00 | 0.01 |
| ORGANIZATION_TYPE_Realtor                | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Religion               | 1.00 | 0.00 |
| ORGANIZATION_TYPE_Restaurant             | 0.00 | 0.01 |
| ORGANIZATION_TYPE_School                 | 0.00 | 0.03 |
| ORGANIZATION_TYPE_Security               | 0.00 | 0.01 |
| ORGANIZATION_TYPE_Security<br>Ministries | 0.00 | 0.01 |
| ORGANIZATION_TYPE_Self-employed          | 0.00 | 0.12 |
| ORGANIZATION_TYPE_Services               | 0.00 | 0.01 |
| ORGANIZATION_TYPE_Telecom                | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Trade: type<br>1       | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Trade: type<br>2       | 0.00 | 0.01 |
| ORGANIZATION_TYPE_Trade: type<br>3       | 0.00 | 0.01 |
| ORGANIZATION_TYPE_Trade: type<br>4       | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Trade: type<br>5       | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Trade: type<br>6       | 0.00 | 0.00 |

|                                     |      |      |
|-------------------------------------|------|------|
| ORGANIZATION_TYPE_Trade: type 7     | 0.00 | 0.03 |
| ORGANIZATION_TYPE_Transport: type 1 | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Transport: type 2 | 0.00 | 0.01 |
| ORGANIZATION_TYPE_Transport: type 3 | 0.00 | 0.00 |
| ORGANIZATION_TYPE_Transport: type 4 | 0.00 | 0.02 |
| ORGANIZATION_TYPE_University        | 0.00 | 0.00 |
| ORGANIZATION_TYPE_XNA               | 0.00 | 0.18 |
| HOUSETYPE_MODE_Missing              | 1.00 | 0.50 |
| HOUSETYPE_MODE_block of flats       | 0.00 | 0.49 |
| HOUSETYPE_MODE_specific housing     | 0.00 | 0.00 |
| HOUSETYPE_MODE_terraced house       | 0.00 | 0.00 |
| WALLSMATERIAL_MODE_Block            | 0.00 | 0.03 |
| WALLSMATERIAL_MODE_Missing          | 1.00 | 0.51 |
| WALLSMATERIAL_MODE_Mixed            | 0.00 | 0.01 |
| WALLSMATERIAL_MODE_Monolithic       | 0.00 | 0.01 |
| WALLSMATERIAL_MODE_Others           | 0.00 | 0.01 |
| WALLSMATERIAL_MODE_Panel            | 0.00 | 0.21 |
| WALLSMATERIAL_MODE_Stone, brick     | 0.00 | 0.21 |
| WALLSMATERIAL_MODE_Wooden           | 0.00 | 0.02 |

|                             |      |      |
|-----------------------------|------|------|
| EMERGENCYSTATE_MODE_Missing | 1.00 | 0.47 |
| EMERGENCYSTATE_MODE_No      | 0.00 | 0.52 |
| EMERGENCYSTATE_MODE_Yes     | 0.00 | 0.01 |